

MODULE 5:

Azure Databases

SQL Azure: Creating a SQL Server - Creating a SQL DB - Creating Tables - Adding Data to the Table - View Connection Strings - Security Configurations - Migrating on premise DB to SQL Azure. Azure Websites: Creating a Website, Setting deployment credentials -Choosing a platform -Setting up Default page for website - Scaling - Auto Scaling by Time -Auto Scaling by Metric - Difference between Free, Shared, Basic and Standard websites - Creating a website using Visual studio.

Part A: SQL Azure Database Management

Assignment 1: Creating and Configuring SQL Azure Database (45 minutes)

Scenario: You are setting up a cloud database for a small e-commerce company called "CloudMart."

Tasks:

1. Create SQL Server in Azure
 - Navigate to Azure Portal and create a new SQL Server
 - Server name: **cloudmart-sql-server-[yourname]**
 - **Location: Choose nearest region**
 - **Authentication: SQL authentication**
 - **Admin username: cloudmartadmin**
 - **Set a strong password and document it securely**
2. Create SQL Database
 - **Database name: CloudMartDB**
 - **Compute + storage: Choose Basic tier (for cost efficiency)**
 - **Backup retention: 7 days**
3. Security Configuration
 - **Add your client IP address to firewall rules**
 - **Create a firewall rule named "OfficeNetwork" with IP range 10.0.0.1-10.0.0.255**
 - **Enable Azure services access to server**
 - **Take screenshots of firewall configuration**

4. Create Tables Connect using Azure Data Studio or SQL Server Management Studio, then execute:

sql

```
CREATE TABLE Products (  
    ProductID INT PRIMARY KEY IDENTITY(1,1),  
    ProductName NVARCHAR(100) NOT NULL,  
    Category NVARCHAR(50),  
    Price DECIMAL(10,2),  
    StockQuantity INT,  
    CreatedDate DATETIME DEFAULT GETDATE()  
);  
  
CREATE TABLE Customers (  
    CustomerID INT PRIMARY KEY IDENTITY(1,1),  
    FirstName NVARCHAR(50),  
    LastName NVARCHAR(50),  
    Email NVARCHAR(100) UNIQUE,  
    Phone NVARCHAR(20),  
    RegistrationDate DATETIME DEFAULT GETDATE()  
);  
  
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY IDENTITY(1,1),  
    CustomerID INT FOREIGN KEY REFERENCES  
Customers(CustomerID),
```

```
OrderDate DATETIME DEFAULT GETDATE(),  
  
TotalAmount DECIMAL(10,2),  
  
Status NVARCHAR(20)  
  
);
```

5. Add Sample Data

sql

```
INSERT INTO Products (ProductName, Category, Price,  
StockQuantity)  
  
VALUES  
  
('Laptop Pro 15', 'Electronics', 1299.99, 50),  
  
('Wireless Mouse', 'Accessories', 29.99, 200),  
  
('USB-C Cable', 'Accessories', 15.99, 500),  
  
('Monitor 27"', 'Electronics', 399.99, 75),  
  
('Keyboard Mechanical', 'Accessories', 89.99, 120);  
  
  
INSERT INTO Customers (FirstName, LastName, Email, Phone)  
  
VALUES  
  
('John', 'Smith', 'john.smith@email.com', '555-0101'),  
  
('Sarah', 'Johnson', 'sarah.j@email.com', '555-0102'),  
  
('Michael', 'Brown', 'mbrown@email.com', '555-0103');  
  
  
INSERT INTO Orders (CustomerID, TotalAmount, Status)  
  
VALUES  
  
(1, 1329.98, 'Completed'),  
  
(2, 489.98, 'Processing'),
```

```
(3, 29.99, 'Shipped');
```

6. Retrieve and Document Connection Strings

- Navigate to your database in Azure Portal
- Click on "Connection strings" under Settings
- Copy and document all four connection string types: ADO.NET, JDBC, ODBC, PHP
- Identify the components: server name, database name, authentication method

7. Query and Verify Execute these queries and document results:

sql

```
-- Verify all data

SELECT * FROM Products;

SELECT * FROM Customers;

SELECT * FROM Orders;


-- Complex query

SELECT c.FirstName, c.LastName, o.OrderID, o.TotalAmount,
o.Status

FROM Customers c

JOIN Orders o ON c.CustomerID = o.CustomerID

ORDER BY o.OrderDate DESC;
```

Deliverables:

- Screenshots of created SQL Server and Database
- Connection strings document (with password masked)
- Screenshots of firewall rules
- Query results screenshots

Assignment 2: Database Migration (30 minutes)

Scenario: Migrate an on-premise database to Azure SQL.

Tasks:

1. Prepare On-Premise Database
 - Install SQL Server Express locally (if not already installed)

- Create a database named **LocalInventoryDB**
- **Create a simple table:**

sql

```
CREATE DATABASE LocalInventoryDB;
```

```
GO
```

```
USE LocalInventoryDB;
```

```
GO
```

```
CREATE TABLE Inventory (
    ItemID INT PRIMARY KEY IDENTITY(1,1),
    ItemName NVARCHAR(100),
    Quantity INT,
    Location NVARCHAR(50)
);
```

```
INSERT INTO Inventory (ItemName, Quantity, Location)
VALUES
```

```
    ('Office Chair', 45, 'Warehouse A'),
```

```
    ('Desk Lamp', 120, 'Warehouse B'),
```

```
    ('Filing Cabinet', 30, 'Warehouse A');
```

2. Migration Methods (Choose one) Option A: Using Azure Data Studio
 - Install Azure Data Studio
 - Install "SQL Server Schema Compare" extension
 - Connect to both local and Azure databases
 - Use Schema Compare to migrate schema
 - Use Import/Export wizard for data
3. Option B: Using BACPAC Export/Import

- Export local database as BACPAC file
 - Upload to Azure Storage (or import directly)
 - Import BACPAC to Azure SQL Database
4. Option C: Using SQL Server Management Studio
 - Right-click on local database
 - Tasks → Deploy Database to Azure SQL Database
 - Follow wizard to complete migration
 5. Post-Migration Verification
 - Verify all tables exist in Azure
 - Verify data integrity
 - Run test queries on Azure database
 - Compare row counts between source and destination

Deliverables:

- Migration method documentation with steps followed
- Screenshots of migration process
- Verification query results
- Any issues encountered and how resolved

Part B: Azure Websites (App Service)

Assignment 3: Creating and Configuring Azure Web App (40 minutes)

Scenario: Deploy a simple web application for CloudMart's marketing site.

Tasks:

1. Create Azure Web App
 - Resource name: **cloudmart-webapp-[yourname]**
 - **Publish: Code**
 - **Runtime stack: Choose one (e.g., .NET 8, Node 18, Python 3.11, PHP 8.2)**
 - **Region: Same as SQL Server**
 - **Pricing plan: Free F1 (for testing)**
2. Set Deployment Credentials
 - **Navigate to Deployment Center**
 - **Create FTP/SCIM credentials**
 - **Username: cloudmart-deploy-user**
 - **Set secure password**
 - **Document credentials safely**
3. Configure Application Settings
 - **Add connection string from SQL Azure Assignment**
 - **Name: CloudMartConnection**
 - **Type: SQLAzure**

- Add application setting: **ENVIRONMENT=Development**
4. Create and Upload Default Page Create **index.html**:

html

```
<!DOCTYPE html>

<html>

<head>

    <title>CloudMart - Welcome</title>

    <style>

        body { font-family: Arial; margin: 40px; background:
#f0f0f0; }

        .container { background: white; padding: 30px;
border-radius: 8px; }

        h1 { color: #0078d4; }

    </style>

</head>

<body>

    <div class="container">

        <h1>Welcome to CloudMart</h1>

        <p>Your Cloud-Based Shopping Platform</p>

        <p>Environment: Development</p>

        <p>Server: Azure App Service</p>

    </div>

</body>

</html>
```

Upload using one method:

- FTP client (FileZilla)
- Azure CLI: **az webapp deployment source config-zip**

- Drag-and-drop in Kudu console (`yourapp.scm.azurewebsites.net`)

5. Verify Deployment

- Browse to `https://cloudmart-webapp-[yourname].azurewebsites.net`
- Take screenshot of live site

Deliverables:

- Screenshots of Web App creation
- Deployment credentials documentation
- Screenshot of live website
- Configuration settings screenshot

Assignment 4: Scaling Configuration (30 minutes)

Tasks:

1. Upgrade to Basic Tier
 - Change from Free to Basic B1
 - Document pricing difference
 - Note additional features unlocked
2. Manual Scaling
 - Configure scale-out to 2 instances
 - Document the process
 - Explain when manual scaling is appropriate
3. Configure Auto-Scaling by Schedule
 - Create rule: "Business Hours Scale-Up"
 - Monday-Friday, 8 AM to 6 PM: 3 instances
 - Other times: 1 instance
 - Take screenshots of configuration
4. Configure Auto-Scaling by Metric
 - Create rule: "CPU-Based Scaling"
 - Scale out: CPU > 70% for 5 minutes, add 1 instance
 - Scale in: CPU < 30% for 10 minutes, remove 1 instance
 - Min instances: 1, Max instances: 4
 - Document configuration
5. Pricing Tier Comparison Create a comparison table documenting:

Feature	Free	Shared	Basic	Standard
Price				

Custom domains
SSL
Auto-scale
Max instances
Daily CPU quota
Backup/Restore
Deployment slots

Deliverables:

- **Screenshots of scaling configurations**
- **Completed comparison table**
- **Analysis of when to use each tier (2-3 paragraphs)**

Assignment 5: Deploy from Visual Studio (45 minutes)

Scenario: Create and deploy a simple web application using Visual Studio.

Tasks:

- 1. Create Project in Visual Studio**
 - Open Visual Studio
 - Create new ASP.NET Core Web App (MVC) or React/Angular app
 - Project name: **CloudMartPortal**
 - **Include sample pages/components**
- 2. Configure Azure Connection**
 - **Right-click project → Publish**
 - **Target: Azure**
 - **Specific target: Azure App Service (Windows or Linux)**
 - **Sign in to Azure account**
 - **Select existing Web App or create new one**
- 3. Add Database Connection Modify `appsettings.json`:**

json

```
{  
  
  "ConnectionStrings": {  
  
    "CloudMartDB": "Your-Azure-SQL-Connection-String"  
  
  },  
  
  "Logging": {  
  
    "LogLevel": {  
  
      "Default": "Information"  
  
    }  
  
  }  
  
}
```

4. Customize Application

- Add page showing data from SQL Azure database
- Create simple product listing page
- Add About page with student info

5. Publish to Azure

- Click Publish button
- Monitor deployment progress
- Verify successful deployment
- Test deployed application

6. Configure CI/CD (Optional Advanced)

- Set up GitHub/Azure DevOps repository
- Configure automated deployment
- Make change and trigger auto-deployment

Deliverables:

- Visual Studio project files (zipped)
- Publish profile screenshots
- Live application URL
- Screenshots of deployed application
- Deployment log/summary

Final Submission Requirements

Create a comprehensive report including:

- 1. Executive Summary (1 page)**
 - Overview of all tasks completed
 - Key learnings
- 2. Detailed Documentation (10-15 pages)**
 - Step-by-step process for each assignment
 - Screenshots at every major step
 - Code snippets used
 - Issues encountered and solutions
- 3. Connection Strings & Credentials (separate secured document)**
 - All connection strings
 - Deployment credentials
 - Access information
- 4. Cost Analysis (1-2 pages)**
 - Azure resources created
 - Estimated monthly costs
 - Recommendations for cost optimization
- 5. Reflection (1 page)**
 - What worked well
 - Challenges faced
 - Real-world applications
 - Future improvements

Submission Format: PDF document + code files in ZIP[IF POSSIBLE]

Estimated Total Time: 4-5 hours

Good luck with your assignment!